

ASSESSMENT TOOL 3 – TECHNICAL PRESENTATION

Course Code:	CSA06	Course Title:	Design and Analysis of Algorithms
CO Assessed:	CO1 – Precision (BL1, BL2, BL3)	Assessment:	Assessment Tool 3 – Technical Presentation
Weightage:	10% of CIA	Total Marks:	100
CO1 Statement:	<i>Determine algorithm efficiency using asymptotic notations and mathematical techniques including Master Theorem and substitution method. (BL1, BL2, BL3)</i>		
Student Name:	B.MANU JASWANTH	Reg. No.:	192525400
Section / Batch:	2025	Date:	27/07/2026

Instructions to Students

- Answer the given question through a technical presentation. Total Marks: 100.
- Clearly define the problem and explain the concept of algorithm growth functions.
- Present comparative analysis using appropriate representations such as graphs, tables, or charts.
- Use suitable tools (PowerPoint, visualization tools, or software) to support your explanation.
- Ensure technical accuracy, logical flow, and proper interpretation of results.
- Maintain clarity in communication, proper slide organization, and professional delivery.
- Time management, confidence, and audience engagement will be considered during evaluation.

Question Type	Focus Area	Questions
Technical Presentation	Analyze and compare growth functions using mathematical techniques and asymptotic concepts and with graphical and visual interpretation	Q1

Code of Conduct

I B.MANU JASWANTH/192525400 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student:MANU

RUBRICS FOR CALCULATION

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Technical Content Accuracy	30%	Highly accurate, in depth, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood
Problem Identification	25%	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach
Use of Tools	20%	Advanced tools, well analyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals
Communication	15%	Highly engaging, confident, and audience focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides

Professionalism	10%	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism; uneven team participation	Lacks professionalism; minimal contribution or ethical awareness
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Marks Evaluation:

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Technical Content Accuracy	30%				
Problem Identification	25%				
Use of Tools	20%				
Communication	15%				
Professionalism	10%				
Total Marks (100):					

Faculty In-charge	Course Coordinator	Head of Department
Name:	Name:	Name:
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____



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Saveetha Institute of Medical And Technical Sciences
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Master Theorem for Non-Linear Recurrence

Presented by

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Problem Statement

- Given recurrence:
- $T(n) = 5T(n/2) + n^2$
- Analyze the recurrence using the Master Theorem.
- Compare $T(n)$ with $n^{\log_2 5}$.
- Identify the applicable Master Theorem case.
- Determine the asymptotic time complexity.
- Validate the result using a recursion tree.



Algorithm Overview



- Divide the problem into **5 equal subproblems**.
- Each subproblem size becomes $n/2$.
- Solve each recursively.
- Merge/combine results with $\diamond\diamond^2$ work.
- Repeat until base case is reached.
- Use Master Theorem for direct complexity analysis.



Algorithm / Pseudocode



Algorithm Solve(n)

if $n == 1$
return 1

return $5 \times \text{Solve}(n/2) + n^2$

Base case: $?? = 1$

Recursive calls: 5

Subproblem size: $??/2$

Merge cost: $??^2$



Complexity Analysis

Compute



Given

$$?? = 5$$

$$?? = 2$$

$$??? = ??^2$$

$$n^{\log_2 5} = n^{\log_2 5}$$

$$\log_2 5 \approx 2.32$$

Compare

$$n^2 = n^{2.32}$$

Since

$$n^2 = n^{2.32 - 0.32}$$

Master Theorem **Case 1** applies.

Final Complexity

$$n^2 = \Theta(n^{2.32 - 0.32})$$

$$\approx \Theta(n^{2.32})$$



Comparative Analysis

Method	Time Complexity
Iterative	Depends on algorithm



Recursion Expansion	$\Theta(n^{2.32})$
Recursion Tree	$\Theta(n^{2.32})$
Master Theorem	$\Theta(n^{2.32})$ (Fastest Analysis)

Observation

- Master Theorem provides the quickest solution.
- Avoids lengthy recurrence expansion.
- Suitable for divide-and-conquer recurrences.



Applications



- Merge-based divide-and-conquer algorithms
- Parallel computing
- Image processing
- Matrix multiplication analysis

- Sorting algorithm analysis
- Recursive optimization algorithms
- Algorithm performance prediction



Results & Discussion

Result

Recurrence analyzed successfully.

Applicable Master Theorem case identified.

Time complexity:

Discussion

Recursive work dominates combine cost.

Number of recursive calls significantly impacts growth.

Complexity is greater than Θ^2 but less than Θ^3 .

Master Theorem simplifies recurrence evaluation.



Conclusion & References



• Conclusion

- Applied Master Theorem to analyze the recurrence.
- Identified **Case 1** as applicable.
- Derived final complexity:
- Demonstrated that Master Theorem is an efficient tool for analyzing divide

and-conquer algorithms. • **References**

- Thomas H. Cormen, *Introduction to Algorithms (CLRS)*, 4th Edition.
- Ellis Horowitz, *Fundamentals of Computer Algorithms*.
- Alfred V. Aho, John E. Hopcroft, Jeffrey D. Ullman, *The Design and Analysis of Computer Algorithms*.

